..... [2]

6 A small coil is positioned so that its axis lies along the axis of a large bar magnet, as shown in Fig. 6.1.

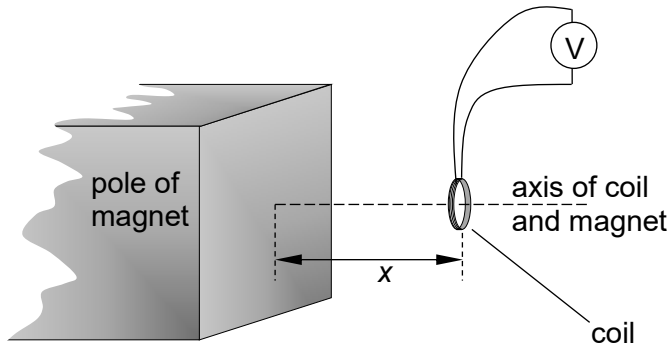

Fig. 6.1

The coil has a diameter of 5.3 mm and contains 180 turns of wire. The ends of the coil are connected to a voltmeter.

The average magnetic flux density *B* through the coil varies with the distance *x* between the face of the magnet and the plane of the coil as shown in Fig. 6.2.

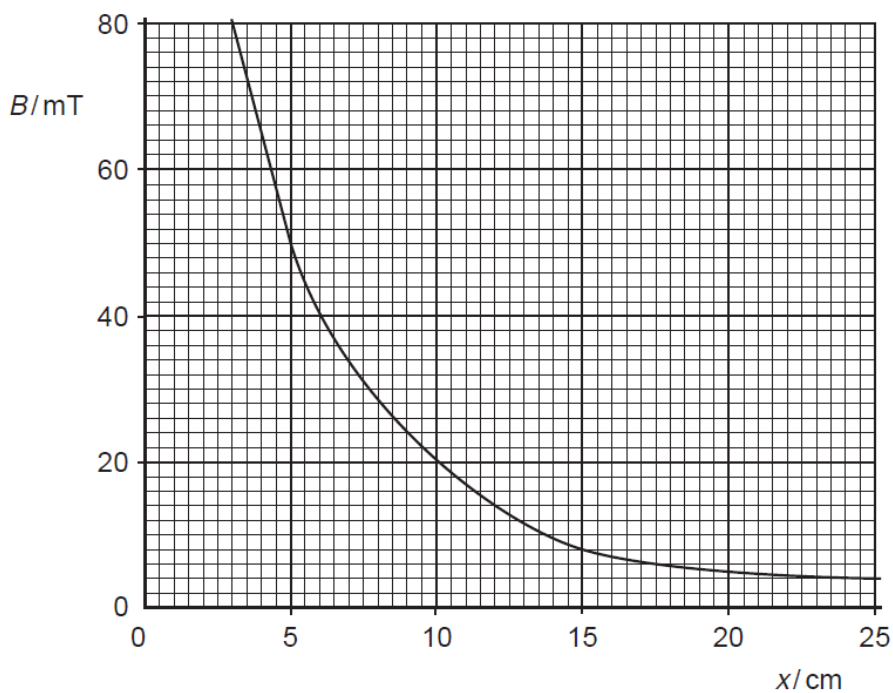

Fig. 6.2

(a) The coil is initially 5.0 cm from the face of the magnet. Show that the magnetic flux linkage of the coil is 2.0×10^{-4} Wb.

[3]

(b) The coil is then moved along the axis of the magnet so that the distance x changes from x = 5.0 cm to x = 20 cm in a time of 0.30 s. As the coil is being moved, a deflection is observed in the voltmeter.

(i) Determine the average electromotive force (e.m.f) induced in the coil.

e.m.f = ……..…….... V [2]

(ii) State and explain the variation, if any, of the speed of the coil so that the induced e.m.f. remains constant during this movement.

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(c) The voltmeter is now replaced with a resistor and the coil is again moved away from the magnet as in **(b)**. As the coil is moved, thermal energy is transferred in the resistor. Use laws of electromagnetic induction to explain the origin of this thermal energy.

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